

Discovery Executive Summary

Project: test-discovery-14aug · Generated: 14/08/2026, 15:12:52

EXECUTIVE SUMMARY

This report consolidates the overall ratings, key findings, and recommended actions from the 1 discovery analysis run across this codebase (frontend and backend). Each section below reproduces that analysis's executive view; full evidence and diagrams live in the individual reports.

Portfolio Overview

#	Analysis	Overall Rating
1	Architecture & Design Analysis	—

1. Architecture & Design Analysis

EXECUTIVE SUMMARY

The Klearcom platform is a monorepo with a Laravel 12 backend, a React 19 SPA frontend, and a Node.js development API. Backend controllers are small in line count (average 74 LOC) but 4 of 6 embed business logic — KPI aggregation, reachability calculations, and report generation — directly against Eloquent models with zero repository abstraction (35 direct ORM access points). Two distinct business algorithms (reachability formula and IVR tree builder) are copy-pasted across 7 locations spanning PHP and JavaScript. The Discovery and Connect domains share three MongoDB collections with no ownership boundary, and cross-domain controllers read models from both domains without an anti-corruption layer. The frontend is architecturally cleaner — React Query and a shared API client handle data access — but two legacy components (one class component with an interval leak, one bypassing React Query) introduce inconsistency. The dominant risk is **change amplification**: modifying the reachability formula or the tree-building algorithm requires synchronized edits across controllers, services, and the dev-API, with no compile-time or test-time guard against drift.

1.1 Benchmark Ratings Summary

#	Hotspot	Primary KPI	Good	Moderate	High Risk	Measured	Rating
H1	Fat Controllers	Avg LOC per controller	<150	150–300	>300	74 LOC	Good
H2	Missing Service Layer	Controller methods accessing models directly	<10	10–20	>20	13	Moderate
H3	Missing Repository Pattern	Direct ORM access points outside repositories	<10	10–20	>20	35 (0 repos)	High Risk
H4	Circular Dependencies	Dependency cycles	0	1–3	>3	0	Good
H5	Shared Utility Abuse	Utility files w/ business logic	0	1–5	>5	1	Moderate
H6	Direct ORM in Controllers	ORM compliance % (queries outside controllers)	>90%	60–90%	<60%	29%	High Risk
H7	God Classes	Classes >1000 LOC	0	1–3	>3	0	Good
H8	Domain Boundary Violations	Cross-domain access points	0	1–5	>5	6	High Risk
H9	Shared Database Coupling	Tables/collections shared across domains	<10%	10–30%	>30%	43% (3/7)	High Risk
H10	Duplicated Business Logic (additional)	Duplicated algorithm instances	0	1–3	>3	7 (2 algorithms)	High Risk
F1	Business Logic in Components	Avg LOC per component	<150	150–300	>300	86 LOC	Good
F2	Missing Frontend Service/Data Layer	Components w/ inline API calls	<10	10–20	>20	2	Good
F3	God / Oversized Components	Components >400 LOC	0	1–3	>3	0	Good
F4	Prop Drilling / Global State Abuse	Max prop-drilling depth	≤2	3–4	>4	2	Good

F5	Legacy / Inconsistent Component Patterns	Legacy-pattern components	0	1–10	>10	2	Moderate
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1.4 Actions Required

Hotspot	Action	Rating	Priority
H3. Missing Repository Pattern	Introduce repository interfaces and Eloquent implementations for all 4 models; bind in AppServiceProvider; replace all 35 direct Model:: calls	High Risk	Critical
H6. Direct ORM in Controllers	Move all 25 Eloquent queries from controllers into repositories; reduce controller methods to validate → service → response	High Risk	Critical
H8. Domain Boundary Violations	Define Discovery and Connect bounded contexts; create published query interfaces; refactor DashboardController and LegacyReportController to use domain interfaces	High Risk	High
H9. Shared Database Coupling	Split 3 shared MongoDB collections into domain-owned collections or enforce per-domain JSON schemas; create domain-specific data-access wrappers	High Risk	High
H10. Duplicated Business Logic	Extract reachability formula into ReachabilityCalculator service; move buildTree() into DiscoveryTreeBuilder; align dev-API with canonical implementations	High Risk	High
H2. Missing Service Layer	Create ConnectService and DiscoveryService; move business logic from 13 controller methods into services	Moderate	High
H5. Shared Utility Abuse	Replace extract() with typed DTOs; move mapping logic from LegacyDataMapper into domain services	Moderate	Medium
F5. Legacy Component Patterns	Rewrite LegacyMonitorPoller as function component with useQuery; rewrite LegacyDashboardWidget with React Query; add ErrorBoundary	Moderate	Medium

1.5 Expected Outcomes

- **Testability without infrastructure:** Repository interfaces allow unit-testing services and controllers with in-memory fakes — no database required.
- **Single-source business logic:** Extracting the reachability calculator and tree builder eliminates 7 duplicated code points, reducing the risk of algorithm drift between production and dev-API.
- **Independent domain evolution:** Bounded contexts for Discovery and Connect allow each module to change its schema, add endpoints, or be extracted into a microservice without breaking the other.
- **Safe schema changes:** Domain-owned MongoDB collections (or enforced schemas) prevent a Discovery-only field change from silently affecting Connect queries or indexes.
- **Consistent frontend patterns:** Converting the 2 legacy components to the React Query + function component standard gives new contributors a single pattern to follow and eliminates the interval-leak and missing-error-boundary risks.


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